
Physics-Guided Temporal Signatures for Data-Scarce Temporal Event Recognition

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Abstract

1 End-to-end sequence models dominate temporal recognition when discriminative
2 structure must be learned from abundant data. We study the complementary
3 regime, in which the event mechanism is known a priori and labelled data are
4 scarce, and formalise a recipe we call *physics-guided temporal signatures* (PGTS):
5 name task primitives from the event mechanism, apply a fixed bank of temporal
6 operators including an order-sensitive asymmetry statistic, select the single split
7 parameter τ on training data only, and train a lightweight classifier. We give three
8 theoretical results: a Bayes-optimal rule and error for the asymmetry statistic under
9 an empirically supported heteroscedastic Gaussian model; an impossibility theorem
10 showing any order-invariant feature map is exactly at chance on permutation-
11 paired temporal-order constructions; and a new finite-sample theorem showing that
12 thresholding the asymmetry statistic separates three-phase events from permuted
13 negatives with failure probability decaying exponentially in sequence length. A
14 controlled benchmark family with three constructions (three-phase falls, sit-to-
15 stand ramps, direction-of-time reversal) confirms the theory: structured signatures
16 and order-sensitive baselines solve all three, order-invariant features sit exactly
17 at 0.500, and sweeps over noise, event timing, duration, length, and distractor
18 channels chart where the inductive bias degrades. As the primary real-data case
19 study, the Dynamic Trajectory Signature (DTS) instantiates PGTS for skeleton-
20 based fall detection: under a strict twin-free protocol, DTS+HGB attains AUROC
21 0.9895 [0.9851, 0.9935] on held-out FallVision, best of thirteen evaluated models,
22 significantly ahead of LSTM/GRU/Transformer, two ST-GCNs, MiniROCKET,
23 and a TCN, and matching the strongest baseline, MultiROCKET (0.9868; paired
24 margin not significant), with $388\times$ fewer features. It leads at every training size,
25 dominates MiniROCKET at every fixed false-positive rate (recall 0.941 vs 0.861
26 at 5% FPR), transfers zero-shot to URFD (0.9200) and GMDCSA-24 (0.9633),
27 matches MultiROCKET under true leave-one-subject-out evaluation (0.941 vs
28 0.934; not significant at four subjects), and recovers the hard Bed fold faster than a
29 matched MiniROCKET baseline at every few-shot budget. All artefacts, manifests,
30 score vectors, seeded benchmark code, and a numbers audit are released.

1 Introduction

32 A central question in machine learning is when task-specific inductive bias, encoded as fixed features,
33 can replace the representational power of learned sequence models. For temporal recognition at
34 scale the answer has been settled in favour of end-to-end learning: spatial-temporal graph networks
35 (Yan, Xiong, and Lin, 2018; Shi et al., 2019; Liu et al., 2020; Chen et al., 2021; Lee et al., 2023)
36 and pretrained pose backbones (Duan et al., 2022) dominate large multi-class skeleton benchmarks,
37 and convolutional transforms (Dempster, Petitjean, and Webb, 2020; Dempster, Schmidt, and Webb,

2021; Tan et al., 2022) and deep architectures (Ismail Fawaz et al., 2020; Bai, Kolter, and Koltun, 2018) dominate generic time-series classification. That dominance rests on two conditions: the discriminative structure is unknown and must be learned, and enough labelled data exist to regularise high-capacity models.

This paper studies the complementary regime, common in safety-critical event detection, where both conditions fail: the event mechanism is formally known, and labelled data are scarce. Our claim is not that a particular application can be solved with handcrafted features; it is a testable claim about inductive bias in data-scarce temporal learning: *when event dynamics are known, a fixed temporal-signature map can remove the burden of learning temporal structure from a generic sequence model*. We formalise the recipe as *physics-guided temporal signatures* (PGTS): (1) name task primitives from the event mechanism; (2) apply a fixed bank of temporal operators, including a temporal-asymmetry statistic $\alpha(z; \tau)$, the fraction of a primitive’s total path variation concentrated in the first τ of the sequence; (3) select τ on training data only; (4) train a lightweight classifier; (5) validate under low-data and domain-shift protocols.

We test this claim along three axes. **Theory.** Theorem 1 derives the Bayes-optimal rule and error for α under an empirically supported heteroscedastic Gaussian model. Theorem 2 proves that any order-invariant feature map is exactly at chance on permutation-paired temporal-order constructions, making temporal ordering provably necessary. Theorem 3 (new relative to prior drafts) upgrades an asymptotic separation argument to a finite-sample guarantee: thresholding α separates three-phase events from permuted negatives with failure probability $\leq 5 \exp(-n\bar{\mu}^2\Delta^2/72B^2)$, with an explicit sample-length condition and a multi-primitive corollary. **Controlled benchmark family.** Three synthetic constructions, three-phase falls, sit-to-stand ramps, and a direction-of-time reversal task, make class-conditional frame multisets identical so only ordering differs; we sweep training-set size, drop SNR, event onset, event duration, sequence length, distractor channels, and operator subsets, comparing the structured signature against order-invariant features and generic order-sensitive learners. **Real-data case study.** The Dynamic Trajectory Signature (DTS), PGTS instantiated for skeleton-based fall detection with eight kinematic primitives and 16 operators (128 features), is evaluated under a strict leakage-corrected protocol against twelve baselines, with low-data learning curves, fixed-false-positive-rate operation, leave-session-out and leave-fall-origin folds, zero-shot transfer to two external datasets, true leave-one-subject-out validation, and an interpretable failure-and-recovery analysis of the hardest fold including a matched-protocol few-shot baseline comparison.

Contributions. (i) The PGTS formalisation and an explicit account of when it should and should not work (Section 6); (ii) three theoretical results, including a new finite-sample separation theorem with explicit constants and a matching impossibility theorem; (iii) an expanded controlled benchmark family in which the theory’s predictions are exactly realised and the boundary of the inductive bias is charted empirically; (iv) a comprehensive, leakage-corrected fall-detection case study in which the 128-dimensional signature with gradient-boosted trees outperforms or matches LSTM/GRU/Transformer, two ST-GCNs, a TCN, MiniROCKET, and MultiROCKET under iid, low-data, fixed-FPR, session, fall-origin, cross-dataset, subject-held-out, and few-shot-adaptation protocols; and (v) a complete reproducibility package with saved score vectors, split manifests, a numbers audit, and seeded benchmark code.

Scope. Real-data superiority claims cover the evaluated baselines only: PoseC3D and other pretrained systems are out of scope because pretraining reintroduces the large-external-data condition our study isolates. FallVision subject IDs are not public, so leave-session-out is used there only as a subject proxy; true leave-one-subject-out evaluation is reported separately on GMDCSA-24, which publishes subject identifiers. Reported runtime covers signature extraction and inference only.

2 Related work

Skeleton-based action recognition and fall detection. ST-GCN (Yan, Xiong, and Lin, 2018) and successors (Shi et al., 2019; Liu et al., 2020; Chen et al., 2021; Lee et al., 2023; Duan et al., 2022) dominate large multi-class skeleton benchmarks; pretrained PoseConv3D-style backbones increasingly support few-shot fine-tuning. PGTS targets the complementary binary, data-scarce regime. Pose-based fall detectors trained on URFD reach 97–98% within-dataset (Lin and Dong, 2021; Zhu et al., 2025); our URFD evaluation is zero-shot transfer. Surveys document device and

environment shift as persistent failure modes (Iguar, Medrano, and Plaza, 2013; Mubashir, Shao, and Seed, 2013; de Miguel et al., 2017).

Time-series classification. ROCKET/MiniROCKET/MultiROCKET (Dempster, Petitjean, and Webb, 2020; Dempster, Schmidt, and Webb, 2021; Tan et al., 2022) are strong generic transforms; InceptionTime (Ismail Fawaz et al., 2020) and TCNs (Bai, Kolter, and Koltun, 2018) are standard deep baselines. We evaluate MiniROCKET, MultiROCKET, and a TCN on the main split, MiniROCKET on the controlled benchmark and under few-shot Bed adaptation, and MultiROCKET under subject-held-out transfer; InceptionTime is released as a configured script.

Inductive bias, sample efficiency, and operating points. Structured inductive biases are a recurring route to sample efficiency (Battaglia et al., 2018), with task-specific features helping in data-scarce medical signal processing (Rajpurkar et al., 2017) and anomaly detection (Pang et al., 2021). Distribution shift and the gap between ranking and calibrated operating points (Guo et al., 2017; Gulrajani and Lopez-Paz, 2021) motivate our fixed-FPR, transfer, and threshold-shift analyses.

3 Theory: asymmetry, impossibility, and finite-sample separation

Definition 1 (Temporal asymmetry operator). *For a scalar primitive trajectory $z = z_{1:T}$ and split $\tau \in (0, 1)$, let $\Delta z_t = z_t - z_{t-1}$ and*

$$\alpha(z; \tau) = \frac{\sum_{t=2}^{\lfloor \tau T \rfloor} |\Delta z_t|}{\sum_{t=2}^T |\Delta z_t| + \epsilon}, \quad \epsilon = 10^{-8}. \quad (1)$$

α is the fraction of total path variation concentrated in the first τ of the sequence: events that concentrate motion late (falls) have small α ; permuted or temporally diffuse signals have $\alpha \approx \tau$.

Theorem 1 (QDA Bayes bound for temporal asymmetry). *Let $\alpha \mid c \sim \mathcal{N}(\mu_c, \sigma_c^2)$, $c \in \{f, n\}$, with equal priors and $\sigma_f \neq \sigma_n$. The Bayes-optimal classifier is a quadratic discriminant rule with two real crossover points $x_1 < x_2$ solving*

$$\left(\frac{1}{\sigma_f^2} - \frac{1}{\sigma_n^2}\right)x^2 - 2\left(\frac{\mu_f}{\sigma_f^2} - \frac{\mu_n}{\sigma_n^2}\right)x + \left(\frac{\mu_f^2}{\sigma_f^2} - \frac{\mu_n^2}{\sigma_n^2}\right) - 2\ln \frac{\sigma_n}{\sigma_f} = 0, \quad (2)$$

with fall assigned to the interior (x_1, x_2) when $\sigma_f < \sigma_n$, and Bayes error $\epsilon_{\text{QDA}}^ = \frac{1}{2}(P(\alpha \notin (x_1, x_2) \mid f) + P(\alpha \in (x_1, x_2) \mid n))$.*

On FallVision’s hip-speed primitive at $\tau^* = 0.30$, $\hat{\mu}_f = 0.171$, $\hat{\sigma}_f = 0.165$, $\hat{\mu}_n = 0.381$, $\hat{\sigma}_n = 0.220$, and Levene’s test rejects equal variance ($p = 1.99 \times 10^{-24}$); (2) gives $x_1 \approx -0.50$ (outside the support of α , so a single threshold $x_2 \approx 0.308$ is active), $P(\text{miss} \mid f) = 0.202$, $P(\text{fa} \mid n) = 0.369$, and $\epsilon_{\text{QDA}}^* = 0.286$. The full eight-primitive signature reduces observed held-out error to 0.051, a factor-5.6 reduction over this single-primitive bound.

Theorem 2 (Order-invariance impossibility). *Call ϕ order-invariant if $\phi(x_{\sigma(1)}, \dots, x_{\sigma(T)}) = \phi(x)$ for every $\sigma \in S_T$. In any construction where each negative is $X^- = \sigma(X^+)$ with $X^+ \sim P_+$ and σ a (random or deterministic) permutation independent of X^+ , every classifier $g = h \circ \phi$ on an order-invariant ϕ has identical class-conditional score laws; hence the ROC curve is the diagonal, AUROC = $\frac{1}{2}$, and the Bayes error is $\frac{1}{2}$.*

Theorem 3 (Finite-sample asymmetry separation). *Model an event primitive by n independent increment magnitudes $w_t \in [0, B]$ with mean μ_s on a still phase $t/n \in [0, p_1)$ and floor phase $[p_2, 1]$, and mean $\mu_d \geq \mu_s$ on the drop phase $[p_1, p_2]$; let the matched negative be a uniformly random permutation of the same increments. Fix $\tau \leq p_1$; set $\bar{\mu} = (p_1 + 1 - p_2)\mu_s + (p_2 - p_1)\mu_d$, $a_f = \tau\mu_s/\bar{\mu}$, and $\Delta = \tau - a_f = \tau(p_2 - p_1)(\mu_d - \mu_s)/\bar{\mu} > 0$. Then for all $n \geq 6/\Delta$, the rule “event iff $\alpha \leq \theta$ ” with $\theta = a_f + \Delta/2$ satisfies*

$$P(\alpha_{\text{event}} \geq \theta) + P(\alpha_{\text{neg}} \leq \theta) \leq 5 \exp\left(-\frac{n\bar{\mu}^2\Delta^2}{72B^2}\right), \quad (3)$$

hence AUROC(α) $\geq 1 - 5 \exp(-n\bar{\mu}^2\Delta^2/72B^2)$ and balanced error $\leq \frac{5}{2} \exp(-n\bar{\mu}^2\Delta^2/72B^2)$.

Corollary 1 (Multiple primitives). *If k primitives independently satisfy Theorem 3 with individual failure probabilities $q < \frac{1}{2}$, majority voting over the single-primitive rules has error at most $\exp(-2k(\frac{1}{2} - q)^2)$.*

Model	Fall		Sit-to-stand		Reversal	
	$n=100$	$n=1000$	$n=100$	$n=1000$	$n=100$	$n=1000$
DTS+HGB (128-d)	0.995 ± 0.002	1.000	0.998 ± 0.002	1.000	1.000	1.000
α -only+LR (8-d)	1.000	1.000	1.000	1.000	1.000	1.000
MiniROCKET+Ridge	1.000	1.000	1.000	1.000	1.000	1.000
Raw-seq MLP	1.000	1.000	1.000	1.000	0.993 ± 0.010	1.000
Bag-of-frames+HGB	0.500	0.500	0.500	0.500	0.500	0.500

Table 1: Expanded controlled temporal-order benchmark: held-out AUROC (mean $\pm$ sd over five seeds, 500 test pairs) on three constructions in which negatives are frame permutations (Fall, Sit-to-stand) or exact time reversals (Reversal) of positives, so class-conditional frame multisets are identical and Theorem 2 applies. Order-invariant bag-of-frames is exactly at chance everywhere; the structured signature and order-sensitive baselines solve all three constructions.

Proofs are in Appendix A; the negative side uses Hoeffding–Serfling concentration for sampling without replacement (Hoeffding, 1963; Serfling, 1974), and the impossibility proof is pointwise in σ , covering deterministic reversal as well as uniform permutation. Three remarks. (a) Theorem 3 makes the sample-length condition explicit: error $\leq \delta$ once $n \geq \frac{72B^2}{\mu^2\Delta^2} \ln \frac{5}{\delta}$, quantifying when a single fixed statistic suffices and no sequence learning is needed. (b) Constants are not optimised; the released verification script confirms the bound and shows empirical failure decaying exponentially with far better constants (e.g., the left side of (3) falls from 0.192 at $n=60$ to 0.003 at $n=500$ at the parameters in Appendix A). (c) A crude capacity heuristic comparing $d = 128$ fixed features to a 216k-parameter LSTM ($n^* = d_{\text{LSTM}} / \ln d_{\text{LSTM}} \approx 17,599$, Appendix B) is reported as intuition only; the low-data claim is supported empirically by the learning curves below, not by the heuristic.

4 Controlled temporal-order benchmark family

The benchmark is synthetic by design: it realises the assumptions of Theorems 2 and 3 exactly, so the theory’s predictions can be checked rather than assumed. Three constructions make class-conditional frame multisets identical so that only temporal ordering carries label information: *fall* (three-phase trajectories: still, drop, floor; negatives are uniform frame permutations), *sit-to-stand* (rise-and-plateau ramps; permuted negatives), and *reversal* (early-burst trajectories; negatives are exact time reversals, isolating direction-of-time information). Each construction uses eight informative channels, pair-level splits, 500 held-out test pairs, and τ selected on training data only by the summed-Welch rule; defaults are $T=150$ and drop SNR 8.

Table 1 reports the headline: DTS-style signatures with gradient boosting and α -only logistic regression solve all three constructions, generic order-sensitive learners (MiniROCKET, an MLP on raw sequences) also succeed, and the order-invariant bag-of-frames classifier is exactly at 0.500 everywhere, as Theorem 2 requires. Appendix Table 8 probes the boundary of the inductive bias with sweeps over drop SNR (1–16), event onset p_1 (0.15–0.7), event duration (0.05–0.4), sequence length (30–300), distractor channels (0–120), and operator subsets. Three findings: (i) order-invariant operator subsets of the signature itself (quantiles, min/max) sit exactly at chance, so the order-sensitive operators carry all discriminative signal; (ii) α -only degrades as SNR approaches 1 (0.879 ± 0.004) and dips mildly under a misspecified fixed τ or very short events, while the full bank and train-selected τ remain at ceiling, consistent with the $\tau \leq p_1$ condition and SNR dependence of Theorem 3; and (iii) at these settings every order-sensitive method, structured or not, solves the constructions even with 120 distractor channels, so the family measures the necessity of temporal order, not separation among order-sensitive methods; the real-data case study below supplies that separation under data scarcity. The original single-construction result (signature AUROC 1.000 at $n=100$; α -only ≥ 0.996 ; bag-of-frames exactly at chance for every n) is reproduced in Appendix Figure 7.

5 Case study: data-scarce skeleton fall detection

Falls have a kinematically distinctive three-phase structure that is formally known rather than latent, and public training data are scarce, making fall detection the natural real-data instantiation of PGTS.

Model	AUROC	F1	Precision	Recall	Accuracy	FP	FN
LSTM	0.9558 [0.9446, 0.9662]	0.905	0.893	0.916	0.900	63	48
GRU	0.9712 [0.9623, 0.9795]	0.920	0.913	0.927	0.917	51	42
Transformer	0.9810 [0.9749, 0.9872]	0.940	0.947	0.934	0.939	30	38
Compact ST-GCN	0.9185 [0.9016, 0.9340]	0.849	0.851	0.847	0.845	85	88
FullST-GCN-COCO	0.9652 [0.9553, 0.9740]	0.907	0.895	0.920	0.903	62	46
MiniROCKET	0.9760 [0.9672, 0.9839]	0.936	0.931	0.941	0.934	40	34
MultiROCKET [†]	0.9868 [0.9808, 0.9920]	0.949	0.954	0.944	0.948	26	32
TCN [†]	0.9629 [0.9523, 0.9734]	0.910	0.893	0.929	0.906	64	41
DTS+LR	0.9433 [0.9297, 0.9559]	0.890	0.883	0.897	0.886	68	59
DTS+ET	0.9801 [0.9734, 0.9861]	0.936	0.907	0.967	0.932	57	19
DTS+RF	0.9785 [0.9711, 0.9849]	0.927	0.893	0.963	0.922	66	21
DTS+HGB	0.9895 [0.9851, 0.9935]	0.951	0.934	0.969	0.949	39	18
DTS-Net	0.9775 [0.9699, 0.9849]	0.928	0.927	0.929	0.926	42	41

Table 2: FallVision held-out results, strict twin-free split (1,115 clips: 574 falls, 541 non-falls). AUROC carries 95% bootstrap intervals; 95% intervals for the remaining metrics are recorded in the released artefacts. Thresholds tuned on validation F1 for every model. [†]Baselines added on identically rebuilt sequences under the same validation-only protocol; harness fidelity: rebuilt DTS features reproduce DTS+HGB’s 0.9895 exactly, and MiniROCKET re-run in this harness gives 0.9797, within its published interval.

The Dynamic Trajectory Signature. Skeleton sequences $S \in \mathbb{R}^{T \times 17 \times 2}$ are hip-centred (midpoint of COCO joints 11–12) and torso-normalised (median shoulder-to-hip distance); missing keypoints are linearly interpolated and keypoints with confidence below 0.10 are zeroed. Eight primitives are extracted per clip (bounding-box width, hip height, torso angle, hip speed, centre-of-mass speed, hip acceleration, shoulder height, head height), each summarised by the 16-operator bank $\Omega(z; \tau) = [\mu, \sigma, \min, \max, Q_{25}, Q_{50}, Q_{75}, R, z_1, z_T, \delta, \beta, \sum |\Delta z|, |\Delta z|_{\max}, |\Delta^2 z|_{\max}, \alpha(z; \tau)]$, yielding $\phi(X) \in \mathbb{R}^{128}$. τ^* maximises the summed two-sample Welch statistic over primitives on the training split (13-point grid on $[0.20, 0.80]$); on FallVision $\tau^* = 0.30$, reused unchanged for external transfer. DTS-Net, an interpretable neural variant (asymmetry-guided attention over the eight blocks; 10,506 parameters), is exploratory; tree ensembles give peak accuracy.

Data, deduplication, and protocol. Parsing the FallVision keypoint archives (FallVision Dataset, 2024) yields 5,793 clips (2,966 falls, 2,827 non-falls) across three acquisition groups (Bed, Chair, Stand) and four sessions. An earlier draft was evaluated on a faulty archive extraction in which clips appeared three to four times and 728 duplicates straddled the train/test boundary, inflating every head-line number;¹ all results here use a strict feature-group twin-free protocol: clips are grouped by feature fingerprint, one representative kept per group (5,572 representatives; 221 duplicate members removed; no conflicting-label groups), then stratified 64/16/20 into 3,565/892/1,115 train/validation/test (574 test falls, 541 non-falls). All model selection uses validation only; thresholds maximise validation F1; AUROC and threshold metrics carry 95% bootstrap intervals from saved score vectors (1,000 resamples, fixed seed); count metrics use Wilson intervals; FP/FN are deterministic counts; the five pre-specified paired-bootstrap comparisons against DTS+HGB remain significant after Bonferroni correction. Baselines: 2-layer LSTM (216,193 parameters), 2-layer GRU, temporal Transformer, Compact ST-GCN (244,161), FullST-GCN-COCO (1,107,905) trained from scratch (Yan, Xiong, and Lin, 2018), MiniROCKET via aeon (Dempster, Schmidt, and Webb, 2021; Middlehurst et al., 2024), MultiROCKET (Tan et al., 2022) (49,728 features; ridge with validation-selected regularisation), a four-block dilated causal TCN (Bai, Kolter, and Koltun, 2018) (43,713 parameters; JAX (Bradbury et al., 2018); Adam, gradient clipping, validation-AUROC checkpointing, 20 epochs), and DTS with LR/ExtraTrees/RF/HGB heads (grids in Appendix Table 6). MultiROCKET and the TCN consume sequences rebuilt from the raw archives via the released manifest; the rebuild reproduces DTS+HGB’s AUROC exactly (0.9895) and MiniROCKET’s within its published interval (0.9797), so the added rows are directly comparable. InceptionTime (Ismail Fawaz et al., 2020) ships as a configured script.

Main result and significance. Table 2: DTS+HGB attains AUROC 0.9895 [0.9851, 0.9935] and F1 0.951 [0.938, 0.963], best of thirteen models. The five pre-specified paired-bootstrap comparisons

¹The earlier draft reported held-out AUROC 0.9998 and recall 1.000 for DTS+HGB; the strict protocol yields 0.9895 and 0.969. The corrected numbers supersede all previously circulated results.

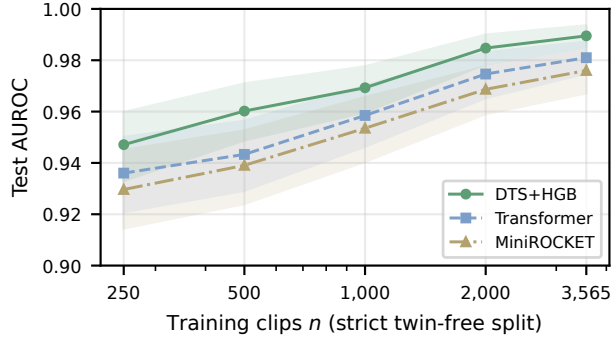


Figure 1: Sample efficiency on the strict twin-free split (shaded bands: 95% bootstrap intervals). DTS+HGB leads at every training size, with the largest margins at the smallest n .

all favour DTS+HGB significantly: +0.0183 [0.0105, 0.0266] vs GRU, +0.0085 [0.0024, 0.0151] vs Transformer, +0.0135 [0.0058, 0.0220] vs MiniROCKET, +0.0710 [0.0568, 0.0879] vs Compact ST-GCN, +0.0242 [0.0166, 0.0338] vs FullST-GCN-COCO. Among the added baselines, DTS+HGB significantly outperforms the TCN (+0.0265 [0.0163, 0.0377]) and *matches* the strongest baseline, MultiROCKET (0.9868 [0.9808, 0.9920]; paired margin +0.0027 [−0.0035, 0.0086], not significant), with 128 interpretable features versus 49,728 and fewer missed falls at the F1 point (18 vs 32). At matched MiniROCKET false positives (40), DTS+HGB detects 557 of 574 falls; at matched recall (548/574), it needs 34 false positives to MiniROCKET’s 47.

Operating points: recall at fixed FPR. Deployment hinges on recall at acceptable alarm rates, since alarm fatigue, not ranking, is the binding constraint in continuous monitoring. DTS+HGB dominates MiniROCKET at every constraint: recall 0.796 [0.761, 0.827] vs 0.629 [0.589, 0.667] at 1% FPR, 0.862 vs 0.777 at 2%, 0.941 [0.918, 0.957] vs 0.861 [0.830, 0.887] at 5%, and 0.981 vs 0.970 at 10%. Against MultiROCKET the picture is honest and mixed: MultiROCKET recalls more falls at the strictest budgets (0.864 vs 0.796 at 1%; 0.882 vs 0.862 at 2%), the methods tie at 5% (0.944 vs 0.941), and DTS+HGB leads at 10% (0.981 vs 0.977) and at the F1 operating point (18 vs 32 missed falls). Operating-point choice between the two is thus a deployment decision at statistically indistinguishable AUROC; what the signature adds at parity is interpretability, a 388× smaller representation, and sub-3 ms CPU inference.

Sample efficiency. Retraining the three strongest families on stratified subsamples ($n \in \{250, \dots, 3,565\}$; validation/test fixed) shows DTS+HGB ahead at every size: 0.9471 vs 0.9360 (Transformer) and 0.9296 (MiniROCKET) at $n=250$, with the largest margins in the scarcest regimes (Figure 1). This sweep, not the capacity heuristic, is the main support for the data-efficiency claim.

Distribution shift: sessions, fall origins, external datasets, subjects. Leave-session-out (subject proxy; FallVision publishes no subject IDs) gives mean AUROC 0.9649 over four folds (Appendix Table 4). Leave-fall-origin gives mean 0.8961: Chair 0.9844 and Stand 0.9916 transfer readily while Bed (0.7123) is the one hard fold, as the QDA analysis predicts (semi-reclined falls shift $\hat{\mu}_f, \hat{\sigma}_f$ outside the Chair+Stand support); the transfer gap is $\Gamma_{F1} = 0.951 - 0.880 = 0.071$ (Appendix Figure 5). Zero-shot transfer of the archived FallVision-trained model: URFD (Kwolek and Kepski, 2014) AUROC 0.9200 [0.8483, 0.9917] with recall 1.000 [0.886, 1.000] (30/30 falls) but 29 false positives, a score-calibration shift rather than a ranking failure (precision 0.508 [0.384, 0.632]; we do not estimate a calibrated threshold without a URFD validation split); GMDCSA-24 (Alam et al., 2024), an uncontrolled-home, low-resolution-webcam dataset, AUROC 0.9633 [0.9362, 0.9849] with the default threshold already usable (recall 0.924, precision 0.880, 10 FP, 6 FN on 160 clips: 79 falls, 81 ADL in our parse). Under *true* leave-one-subject-out on GMDCSA-24 (published subject IDs; four subjects), DTS+HGB attains mean AUROC 0.941 versus MultiROCKET 0.934, MiniROCKET 0.918, and α -only 0.917 (Table 3); the DTS–MultiROCKET margin (paired pooled out-of-fold AUROC +0.012 [−0.034, 0.056]) is not significant at four subjects, so we state the result as a match:

Model	S1	S2	S3	S4	Mean
DTS+HGB	0.953	0.969	0.970	0.871	0.941
MultiROCKET	0.988	0.882	0.948	0.918	0.934
MiniROCKET	0.988	0.816	0.942	0.926	0.918
α -only+LR	0.977	0.944	0.965	0.782	0.917

Table 3: True leave-one-subject-out AUROC on GMDCSA-24 (4 subjects). DTS+HGB matches the modern MultiROCKET baseline under person-level shift, with a small, non-significant mean advantage and zero sequence-learning parameters.

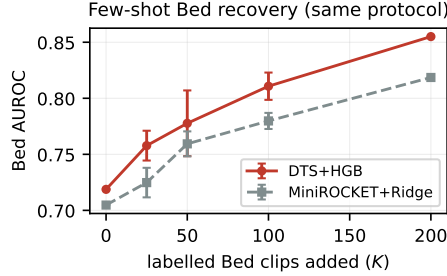


Figure 2: Few-shot Bed recovery under the identical protocol for both models (mean $\pm$ 95% interval over five seeds; fixed 1,547-clip Bed test). DTS+HGB recovers faster and higher than MiniROCKET+ridge at every adaptation budget.

241 a fixed 128-dimensional signature with zero sequence-learning parameters matches a state-of-the-art
242 convolutional model under person-level shift.

243 **Failure analysis and few-shot recovery.** The Bed fold is characterised rather than hidden: it is a
244 representation shift, not miscalibration (AUROC is threshold-free and re-tuning the Bed threshold
245 barely moves F1). Missed Bed falls have higher hip-speed asymmetry ($\alpha \approx 0.33$) than detected
246 ones ($\alpha \approx 0.21$), exactly the direction the QDA model predicts for semi-reclined, less front-loaded
247 descents. Adding K labelled Bed clips to training lifts Bed AUROC monotonically from 0.714 ($K=0$)
248 to 0.784 ($K=50$) and 0.848 ($K=200$), reproduced by an independent five-seed re-implementation
249 (Appendix Figure 6; the recovery protocol reserves part of the Bed fold as the adaptation pool, so its
250 zero-shot start differs slightly from the full-fold 0.7123); the fall/non-fall score separation widens
251 from 0.45 to 0.59. Under the identical protocol (same base, adaptation pool, and fixed 1,547-clip Bed
252 test; five seeds), DTS+HGB dominates a MiniROCKET+ridge baseline at every adaptation budget,
253 from 0.719 vs 0.705 zero-shot to 0.855 vs 0.819 at $K=200$ (Figure 2): the structured representation
254 also adapts more sample-efficiently. The hard fold reveals the limits of the inductive bias; the failure
255 is interpretable, predicted by the theory, and largely closed with minimal target supervision. Figure 3
256 visualises the calibration story from the saved score vectors: FallVision and GMDCSA-24 are cleanly
257 separated at the default threshold (non-fall mass above 0.5: 7.2% and 12.3%), while the Bed fold’s
258 high-score non-fall mass is partially cleared by adaptation; URFD vectors were not retained from the
259 archived run, so its calibration shift is quantified in the text only.

260 **Ablations and feature analysis.** Single-family removal changes AUROC by at most -0.0039
261 (Appendix Table 7): the signature is intentionally over-complete, with partially redundant families
262 providing robustness to missing or noisy keypoints. Zeroing all eight α coordinates costs only
263 -0.0023 AUROC on real data; this does not show α is unnecessary, because other operators (slopes,
264 velocities, accelerations) encode correlated order information, and the controlled benchmark, where
265 marginal information is identical by construction, isolates the pure ordering contribution that the
266 real-data ablation cannot (order-invariant operator subsets drop to chance there, Appendix Table 8).
267 DTS-Net attention correlates positively but non-significantly with tree importances ($r = 0.650$,
268 $p = 0.081$, $k = 8$ families) and is treated as exploratory; across all 128 coordinates, tree-importance
269 ranks correlate with label-association ranks ($\rho = 0.523$ and $\rho = 0.704$ against Spearman and Welch
270 statistics), and seven of the top twelve features are asymmetry coordinates. Extraction plus inference
271 run in under 3 ms per clip on commodity CPU (<50 MB), excluding pose estimation.

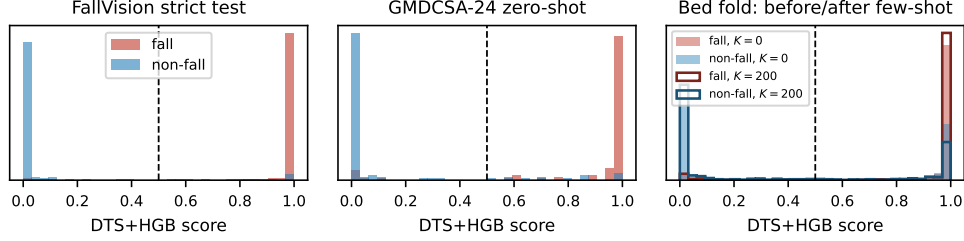


Figure 3: Class-conditional DTS+HGB score distributions from the saved score vectors (dashed line: fixed threshold 0.5). Left: FallVision strict test. Centre: GMDCSA-24 zero-shot (default threshold usable). Right: the hard Bed fold before ($K=0$, filled) and after ($K=200$, outlined) few-shot adaptation, which thins the high-score non-fall mass and widens the fall/non-fall separation from 0.44 to 0.59.

6 When should physics-guided temporal signatures work?

The theory and the two experimental regimes support an explicit scope statement. PGTS should help when: (1) the event mechanism is known, so primitives can be named a priori; (2) the discriminative signal is low-dimensional relative to the raw sequence; (3) labelled data are scarce relative to the capacity of competitive sequence models; (4) temporal order carries necessary information (Theorem 2 fails for order-invariant maps precisely then); and (5) domain shift acts interpretably through primitive distributions, as in the Bed fold, enabling diagnosis and few-shot repair. PGTS should be expected to fail or lose when: event structure is unknown or heterogeneous; many classes differ subtly; appearance or context cues dominate kinematics; abundant labelled data or pretrained models restore the advantage of learned representations; or the operator bank is misaligned with event timing, the failure mode realised in the benchmark when a fixed τ violates $\tau \leq p_1$ and the SNR sweep approaches 1. These conditions are testable in advance of training, which is what makes the recipe useful rather than merely retrospective.

7 Limitations

Real-data evidence is strongest for skeleton kinematics and binary event detection; we make no claim for large multi-class recognition. FallVision lacks public subject IDs (leave-session-out is a proxy; the leakage erratum and corrected protocol are described in Section 5); GMDCSA-24 LOSO has only four subjects, so the MultiROCKET comparison is underpowered and is stated as a match. PoseC3D and pretrained or fine-tuned skeleton-action systems are not evaluated; pretraining reintroduces large external data and may dominate, so evaluated-baseline superiority is the only claim. Bed-origin falls remain hardest (AUROC 0.7123 zero-shot) and recovery requires labelled target clips; URFD transfer requires threshold calibration. Subjects are healthy adults aged 22–27; elderly and frail-adult falls may violate the three-phase pattern more often. PGTS assumes known event mechanics and a low-dimensional discriminative signal; the finite-sample theorem idealises increments as independent and bounded. Runtime excludes pose estimation; DTS-Net attention is directional evidence, not validated explanation. Tree ensembles may be less natural than neural models for some end-to-end deployments.

8 Conclusion

When event dynamics are known and data are scarce, a fixed physics-guided temporal signature with a lightweight classifier is a theoretically grounded, empirically strong, and interpretable alternative to generic sequence learning. The impossibility and finite-sample theorems delimit exactly when temporal order must be modelled and when a single fixed statistic provably suffices; the controlled benchmark family realises both predictions; and the fall-detection case study shows the recipe surviving strict leakage-proof evaluation, low-data regimes, fixed-false-positive-rate operation, and person-level distribution shift against strong learned baselines. We release all artefacts for one-command reproduction.

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376 A Proofs

377 **Theorem 2.** By order-invariance, $\phi(\sigma(x)) = \phi(x)$ for every $\sigma \in S_T$ and every x . A negative is
 378 $X^- = \sigma(X^+)$ with $X^+ \sim P_+$ and σ independent of X^+ . Pointwise, $g(X^-) = h(\phi(\sigma(X^+))) =$
 379 $h(\phi(X^+)) = g(X^+)$, so the class-conditional score laws coincide. For any threshold t , $\text{TPR}(t) =$
 380 $\text{FPR}(t)$, so the ROC curve is the diagonal, $\text{AUROC} = \frac{1}{2}$, and the Bayes classifier on ϕ cannot beat
 381 the prior. The argument is pointwise in σ , so it covers deterministic permutations such as exact time
 382 reversal. $\square$

383 **Theorem 3.** Write $m = \lfloor \tau n \rfloor$, $S_\tau = \sum_{t \leq m} w_t$, $S = \sum_{t \leq n} w_t$, $\alpha = S_\tau/S$ (the stabiliser ϵ only
 384 decreases α and is absorbed). All $w_t \in [0, \bar{B}]$. Fix $\epsilon = \bar{\mu}\Delta/12 \leq \bar{\mu}/2$.

385 *Event side.* Since $\tau \leq p_1$, the first m increments have mean μ_s , so $E[S_\tau] \leq \tau n \mu_s$ and $E[S] = n \bar{\mu}$.
 386 By Hoeffding (Hoeffding, 1963), $P(S_\tau \geq n(\tau \mu_s + \epsilon)) \leq \exp(-2\epsilon^2 n^2 / (m B^2)) \leq \exp(-2\epsilon^2 n / B^2)$
 387 and $P(S \leq n(\bar{\mu} - \epsilon)) \leq \exp(-2\epsilon^2 n / B^2)$. On the complement, using $(\bar{\mu} - \epsilon)^{-1} \leq (1 + 2\epsilon/\bar{\mu})/\bar{\mu}$
 388 for $\epsilon \leq \bar{\mu}/2$,

$$\alpha_{\text{event}} \leq \frac{\tau \mu_s + \epsilon}{\bar{\mu} - \epsilon} \leq a_f + \frac{2\epsilon\tau}{\bar{\mu}} + \frac{2\epsilon}{\bar{\mu}} \leq a_f + \frac{4\epsilon}{\bar{\mu}} = a_f + \frac{\Delta}{3} < \theta.$$

389 *Negative side.* Conditional on the realised multiset $\{w_t\}$, the first m entries of the permuted sequence
 390 are a uniform sample without replacement, so by the Hoeffding–Serfling inequality (Hoeffding, 1963;
 391 Serfling, 1974), $P(|S_\tau^{\text{neg}} - \frac{m}{n} S| \geq \epsilon n \mid \{w_t\}) \leq 2 \exp(-2\epsilon^2 n^2 / (m B^2)) \leq 2 \exp(-2\epsilon^2 n / B^2)$.
 392 Combined with the lower-tail bound on S and $m/n \geq \tau - 1/n$, on the complement,

$$\alpha_{\text{neg}} \geq \frac{m}{n} - \frac{\epsilon n}{S} \geq \tau - \frac{1}{n} - \frac{2\epsilon}{\bar{\mu}} \geq \tau - \frac{\Delta}{6} - \frac{\Delta}{6} = \tau - \frac{\Delta}{3} > \theta,$$

393 using $n \geq 6/\Delta$ and $\tau - \Delta/3 > a_f + \Delta/2 \iff \Delta/6 > 0$. Summing the five failure probabilities
 394 with $\epsilon = \bar{\mu}\Delta/12$ gives (3); the AUROC and balanced-error consequences follow as in the main text.
 395 Corollary 1 is Hoeffding’s inequality applied to the k vote indicators. $\square$

396 *Numerical verification.* With $\mu_s=0.1$, $\mu_d=1.0$, $p_1=0.3$, $p_2=0.5$, $\tau=0.25$, increments uniform on
 397 $[0, 2\mu]$ ($B=2$), 20,000 trials per n : the empirical left side of (3) is 0.192 ($n=60$), 0.100 ($n=120$),
 398 0.030 ($n=250$), 0.003 ($n=500$), 0.000 ($n=1000$), never violating the bound, with empirical AUROC
 399 0.976, 0.999, 1.000, 1.000, 1.000 respectively (verify_theorem.py).

400 B Capacity heuristic (intuition only)

401 Take $d_{\text{DTS}} = 128$ and, as a crude capacity proxy, $d_{\text{LSTM}} = 216,193$ (trainable parameters of the
 402 2-layer LSTM baseline). The scale $n^* = d_{\text{LSTM}} / \ln d_{\text{LSTM}} \approx 17,599$ marks where crude $\sqrt{d/n}$
 403 arguments favour the low-dimensional representation; the curves are parallel on log-log axes and
 404 never cross (Figure 4), the derivation omits constants, and FallVision’s $n_{\text{train}} = 3,565$ is $4.9\times$ below
 405 n^* . This is intuition, not a bound; the learning curves in Figure 1 are the evidence.

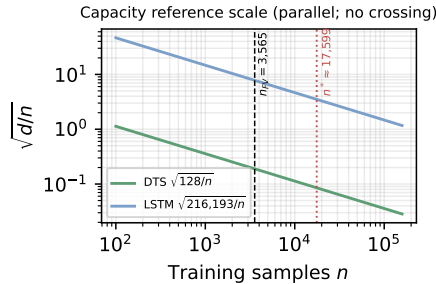


Figure 4: Heuristic capacity comparison: parallel $\sqrt{d/n}$ curves; n^* (dotted) is a derived scale, not a crossing point; $n_{\text{train}} = 3,565$ (dashed).

Session	n	Falls	F1	AUROC	Recall
1	1,513	679	0.866 [0.847, 0.884]	0.9382 [0.9259, 0.9492]	0.944
2	1,593	1,072	0.966 [0.958, 0.973]	0.9926 [0.9898, 0.9951]	0.954
3	2,064	903	0.925 [0.912, 0.937]	0.9840 [0.9797, 0.9879]	0.925
4	402	213	0.868 [0.836, 0.900]	0.9448 [0.9227, 0.9645]	0.953
Mean	–	–	0.906	0.9649	–

Table 4: Leave-session-out (subject proxy) for DTS+HGB, fixed threshold 0.5; 95% bootstrap intervals.

Held-out	n	Falls	F1	AUROC
Bed	1,747	922	0.768 [0.749, 0.788]	0.7123 [0.6868, 0.7357]
Chair	1,878	961	0.934 [0.923, 0.944]	0.9844 [0.9799, 0.9883]
Stand	1,947	984	0.937 [0.926, 0.949]	0.9916 [0.9886, 0.9945]
Mean	–	–	0.880	0.8961

Table 5: Leave-fall-origin robustness for DTS+HGB, fixed threshold 0.5; 95% bootstrap intervals.

C Additional real-data tables

D Original Bed-recovery reproduction

E Controlled-benchmark details and sweeps

Construction defaults: $T=150$, eight informative channels, drop SNR 8, onset $p_1=0.5$ (fall) / 0.4 (sit-to-stand), duration 0.2 (fall) / 0.3 (sit-to-stand), 500 held-out test pairs per run; τ selected on training data only by the summed-Welch rule; bag-of-frames uses per-channel value quantiles (order-invariant by construction). All runs, seeds, and per-run results are in `results_all.jsonl`; `synth_bench.py` regenerates everything and `analyze_bench.py` regenerates the tables.

F Reproducibility

Code, extracted features, split manifests, the deduplication audit, saved score vectors, hyperparameter grids, selected configurations, per-model thresholds, a numbers audit covering every headline value, and seeded one-command reproduction scripts (controlled-benchmark suite, sweep aggregation, theorem verification) are provided in the supplementary material and will be released publicly upon publication.

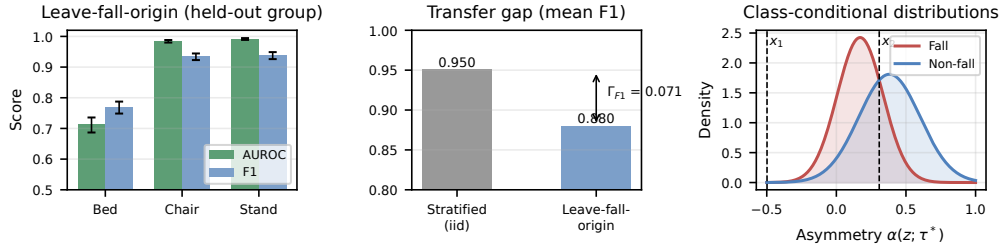


Figure 5: Left: leave-fall-origin AUROC and F1 with 95% intervals. Centre: transfer gap versus the stratified iid reference (reference shown to two decimals in the panel; $\Gamma_{F1} = 0.951 - 0.880 = 0.071$). Right: class-conditional hip-speed asymmetry distributions with QDA boundaries; only $x_2 \approx 0.308$ is active since $\alpha \geq 0$.

Model	Search space	Selected
DTS+LR	$C \in \{0.05, 0.1, 0.5, 1, 2, 10\}$	$C=0.05$
DTS+ET	trees $\{100, 300, 600\} \times \text{feat. } \{\sqrt{d}, 0.25\}$	100 trees, max_features=0.25
DTS+RF	trees $\{100, 300, 600\} \times \text{feat. } \{\sqrt{d}, 0.25\}$	600 trees, max_features=0.25
DTS+HGB	iter. $\{100, 300, 600\} \times \text{lr } \{0.05, 0.1, 0.2\} \times \lambda_2 \{0, 0.1\}$	600 iter., lr=0.1, $\lambda_2=0.1$
MiniROCKET	kernels $\{1000, 5000, 10000\}$	10,000

Table 6: Validation-only hyperparameter grids and selected configurations.

Configuration	F1	AUROC	$\Delta F1$	ΔAUC
Full model	0.951	0.9895	—	—
–BBox-W	0.942	0.9882	−0.009	−0.0013
–Hip-Y	0.939	0.9855	−0.012	−0.0039
–Torso-Ang	0.945	0.9892	−0.006	−0.0003
–Hip-Spd	0.949	0.9897	−0.002	+0.0002
–Ctr-Spd	0.949	0.9904	−0.002	+0.0009
–Hip-Acc	0.947	0.9894	−0.004	−0.0001
–Shldr-Y	0.943	0.9882	−0.008	−0.0012
–Head-Y	0.951	0.9903	−0.001	+0.0008
Asymmetry zeroed ($\alpha=0$)	0.939	0.9872	−0.012	−0.0023

Table 7: DTS+HGB feature ablation, main protocol (each row retrained; threshold re-tuned on validation).

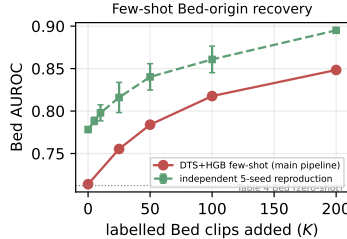


Figure 6: Original few-shot Bed-origin recovery: main-pipeline curve and an independent five-seed re-implementation (95% intervals); both recover the fold monotonically. The matched-protocol baseline comparison is in main-text Figure 2.

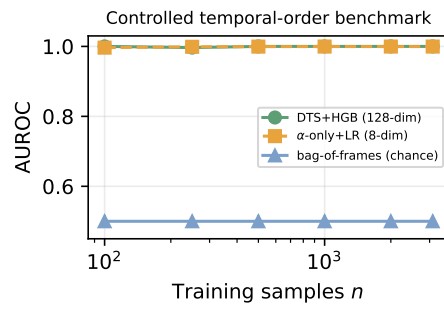


Figure 7: Original single-construction benchmark: structured signature and α -only saturate; order-invariant bag-of-frames sits exactly at chance for every n (Theorem 2).

Setting	DTS+HGB	α -only+LR	Bag-of-frames	MiniROCKET
<i>Drop SNR μ_d/σ_s (fall, $n=250$)</i>				
SNR 1	1.000	0.879 ± 0.004	0.500	1.000
SNR 2	1.000	0.993 ± 0.001	0.500	1.000
SNR 4	1.000	1.000	0.500	1.000
SNR 8	1.000	1.000	0.500	1.000
SNR 16	1.000	1.000	0.500	1.000
<i>Event onset p_1 (fall, $n=250$)</i>				
$p_1=0.15$	1.000	1.000	0.500	–
$p_1=0.3$	1.000	1.000	0.500	–
$p_1=0.5$	1.000	1.000	0.500	–
$p_1=0.7$	1.000	1.000	0.500	–
<i>α-only with misspecified fixed $\tau=0.30$ vs train-selected τ</i>				
$p_1=0.15$	selected: 1.000		fixed 0.30: 1.000	
$p_1=0.3$	selected: 1.000		fixed 0.30: 0.993 ± 0.001	
$p_1=0.5$	selected: 1.000		fixed 0.30: 1.000	
$p_1=0.7$	selected: 1.000		fixed 0.30: 0.999	
<i>Distractor channels added (fall, $n=250$)</i>				
+0 noise ch.	1.000	1.000	0.500	1.000
+8 noise ch.	1.000	1.000	0.500	1.000
+32 noise ch.	1.000	1.000	0.500	1.000
+120 noise ch.	1.000	1.000	0.500	1.000
<i>Event duration p_2-p_1 (fall, $n=250$)</i>				
dur 0.05	1.000	0.992 ± 0.001	0.500	
dur 0.1	1.000	0.999	0.500	
dur 0.2	1.000	1.000	0.500	
dur 0.4	1.000	1.000	0.500	
<i>Sequence length T (fall, $n=250$)</i>				
$T=30$	1.000	0.999	0.500	
$T=60$	1.000	1.000	0.500	
$T=150$	1.000	1.000	0.500	
$T=300$	1.000	1.000	0.500	
<i>Operator-bank ablation, DTS+HGB (fall, $n=250$)</i>				
full 16-operator bank			1.000	
– all α			1.000	
α only			1.000	
order-sensitive ops only			1.000	
order-invariant ops only			0.500	
quantiles/min/max only			0.500	
slopes only			1.000	
endpoints only			0.997 ± 0.002	

Table 8: Controlled-benchmark sweeps (held-out AUROC, mean $\pm$ sd over three seeds, 500 test pairs). Bag-of-frames remains at exactly 0.500 in every setting, as Theorem 2 requires, and order-invariant operator subsets of the signature itself (quantiles, min/max) also sit exactly at chance, confirming that the order-sensitive operators carry all discriminative signal. α -only degrades as drop SNR approaches 1 and dips mildly under a misspecified fixed τ or very short events, while the full operator bank and train-selected τ remain at ceiling. At these settings every order-sensitive method solves the constructions, including with 120 distractor channels: the family measures the necessity of order sensitivity, not separation among order-sensitive methods.

NeurIPS Paper Checklist

1. **Claims.** Do the main claims made in the abstract and introduction accurately reflect the paper’s contributions and scope?
[Yes] Claims are stated per-protocol with confidence intervals; superiority claims are explicitly restricted to evaluated baselines, and the LOSO comparison is stated as a match, not a win.
2. **Limitations.** Does the paper discuss the limitations of the work?
[Yes] Section 7 covers dataset, population, baseline-scope, theoretical-assumption, and deployment limitations, including a corrected-leakage erratum.
3. **Theory assumptions and proofs.** For each theoretical result, does the paper provide the full set of assumptions and a complete (and correct) proof?
[Yes] Assumptions are stated in each theorem; complete proofs are in Appendix A with a numerical verification script.
4. **Experimental result reproducibility.** Does the paper fully disclose all the information needed to reproduce the main experimental results?
[Yes] Protocols, splits, grids, thresholds, and seeds are specified; the benchmark is fully regenerable from released seeded code; real-data score vectors and manifests are in the supplementary artefacts.
5. **Open access to data and code.** Does the paper provide open access to the data and code?
[Yes] All datasets are public (FallVision, URFD, GMDCSA-24); code and artefacts are in the supplementary material and will be released publicly.
6. **Experimental setting/details.** Does the paper specify all the training and test details?
[Yes] Section 5 and Appendices C–D.
7. **Experiment statistical significance.** Does the paper report error bars or other appropriate information about statistical significance?
[Yes] 95% bootstrap/Wilson intervals on every scalar; pre-specified paired-bootstrap comparisons with Bonferroni correction; multi-seed means and standard deviations on the benchmark.
8. **Experiments compute resources.** Does the paper provide sufficient information on the computer resources needed?
[Yes] All models train on commodity CPU/GPU at small scale; signature extraction and inference run in under 3 ms per clip on CPU; the benchmark suite runs in under one CPU-hour.
9. **Code of ethics.** Does the research conform with the NeurIPS Code of Ethics?
[Yes]
10. **Broader impacts.** Does the paper discuss both potential positive and negative societal impacts?
[Yes] Fall detection supports ageing-in-place care; risks discussed include false-alarm fatigue, missed falls at miscalibrated thresholds, and demographic shift from healthy-adult training data (Sections 5–7).
11. **Safeguards.** Does the paper describe safeguards for responsible release?
[NA] No high-misuse-risk models or data are released.
12. **Licenses for existing assets.** Are the creators or original owners of assets used in the paper properly credited, and are the license and terms of use explicitly mentioned and properly respected?
[Yes] FallVision (Harvard Dataverse), URFD, GMDCSA-24, and aeon are cited with their published sources.
13. **New assets.** Are new assets introduced in the paper well documented?
[Yes] The benchmark generators, audit, and reproduction scripts are documented in the supplementary material.
14. **Crowdsourcing and research with human subjects.**
[NA] Only existing public datasets are used.
15. **IRB approvals.**
[NA] No new human-subjects data were collected.